



PROJECT VIKSIT BHARAT

20
26

A NATIONAL LEVEL INNOVATION HACKATHON

Innovate for **India**. Build for **Tomorrow**.



Team Name :Quantum Quotients

Team Leader Name : Golla Bhargava Teja

Problem Statement : Modern cities generate vast amounts of data through surveillance networks, traffic systems, emergency services, and public infrastructure. However, these systems often operate in isolation, leading to delayed responses, fragmented decision-making, and inefficient incident management. As urban environments become increasingly complex, there is a growing need for a unified intelligence platform that can observe, understand, predict, coordinate, and respond to city-wide challenges in real time.

Team Members

Team Leader: Golla Bhargava Teja

Team Member-1:

Name: Golla Bhargava Teja

Name: Koduri Shripaad

College: CMR College of Engineering And
Technology

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ABSTRACT & PROBLEM STATEMENT

Problem Statement:

- ❑ Modern cities generate millions of data points every day through **CCTV** networks, traffic systems, emergency services, public infrastructure, and citizen interactions. Yet these systems operate independently, creating fragmented awareness and delayed response during critical situations.
- ❑ As a result, cities often react **after** incidents occur rather than preventing or coordinating them in real time.

Project Objectives

- ❑ **Laminar** is designed as an **AI Operating System for Cities** that transforms passive infrastructure into an intelligent decision-making network capable of:
 - ✓ Observing city-wide activities in real time
 - ✓ Understanding and analyzing complex urban situations
 - ✓ Predicting risks before escalation
 - ✓ Coordinating emergency response across systems
 - ✓ Reconstructing incidents for accountability and investigation

Existing Challenges:



Emergency Response Delays

Ambulances lose valuable time in traffic congestion.
Emergency incidents are often detected too late.



Crowd & Public Safety Risks

Stampedes, overcrowding, panic situations, and uncontrolled gatherings can escalate rapidly.
Manual monitoring cannot continuously predict crowd behavior.



Environmental & Infrastructure Threats

Floods, waterlogging, and urban disruptions impact mobility and public safety.
Authorities often lack real-time situational awareness.



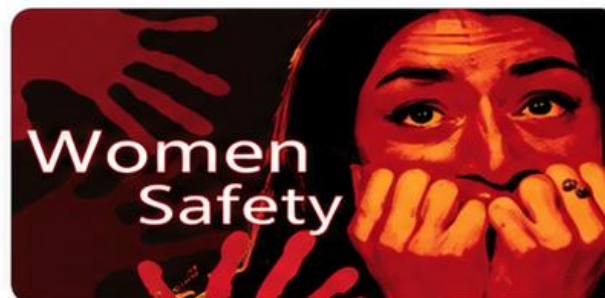
Citizen Safety Concerns

Women, students, senior citizens, and vulnerable individuals may face emergencies without immediate assistance.
Existing safety systems depend heavily on manual reporting.



Post-Incident Investigation Complexity

Investigators spend hours reviewing surveillance footage.
Evidence remains scattered across multiple disconnected systems.





PROPOSED SOLUTION

Solution Overview:

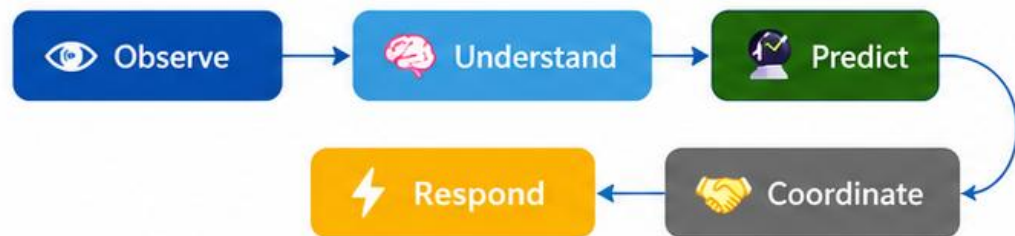
Laminar is a unified AI-powered city intelligence platform that transforms passive infrastructure into an active decision-making network. It continuously monitors, **analyzes**, predicts, and coordinates responses across public safety, mobility, emergency management, and urban operations in real time.

Why Laminar is Different?

Traditional City Systems



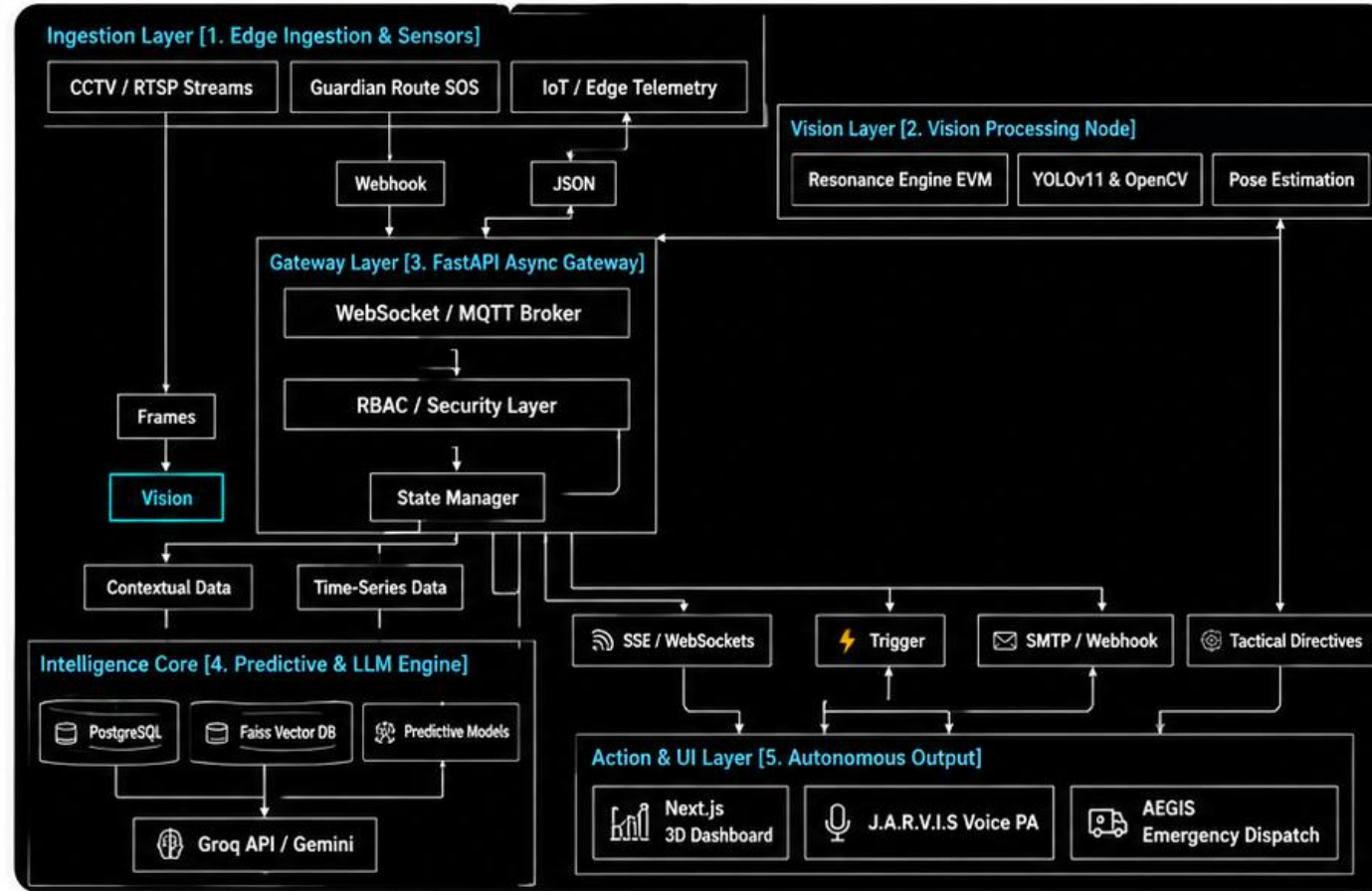
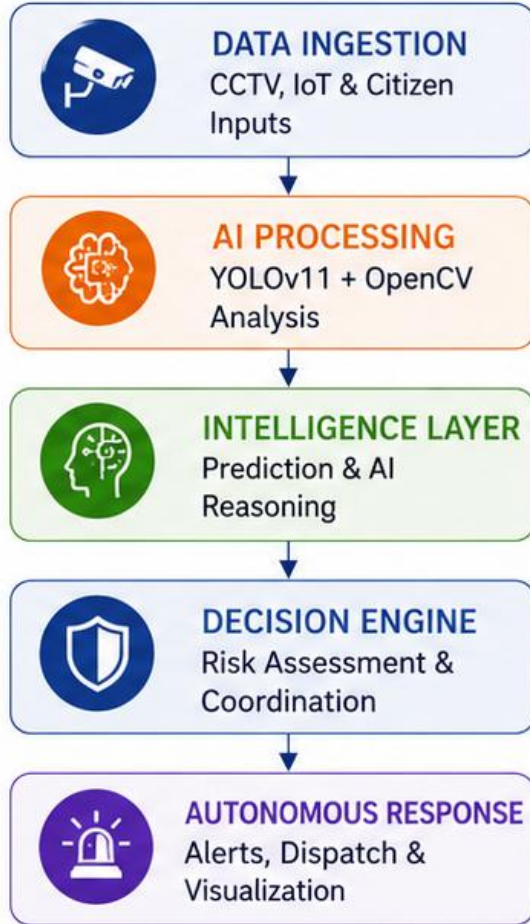
Laminar:



Urban Challenge	AI Response System
Crowd Congestion & Stampede Risk	 Crowd Intelligence
Women Safety Threats	 Guardian Route
Missing Persons Cases	 Amber Protocol
Emergency Vehicle Delays	 Green Wave
Flood & Waterlogging Events	 Flood Intelligence
Medical Emergencies	 AEGIS Protocol
Citizen Distress Events	 Sentinel Beacon
CCTV Investigation Delays	 AI Video Search
Incident Reconstruction	 Liability + 4D Playback
Critical Infrastructure Monitoring	 Resonance Bridge Intelligence
Natural Language Intelligence	 Randy AI



METHODOLOGY / WORKFLOW



From Raw Data to Autonomous Action



SYSTEM ARCHITECTURE



Asynchronous Processing

FastAPI + WebSockets for real-time updates



AI-Native Intelligence

YOLOv11, OpenCV, Gemini & Predictive Models



Hybrid Data Storage

PostgreSQL + Faiss Vector Database



Secure & Scalable

JWT Authentication + RBAC Access Control

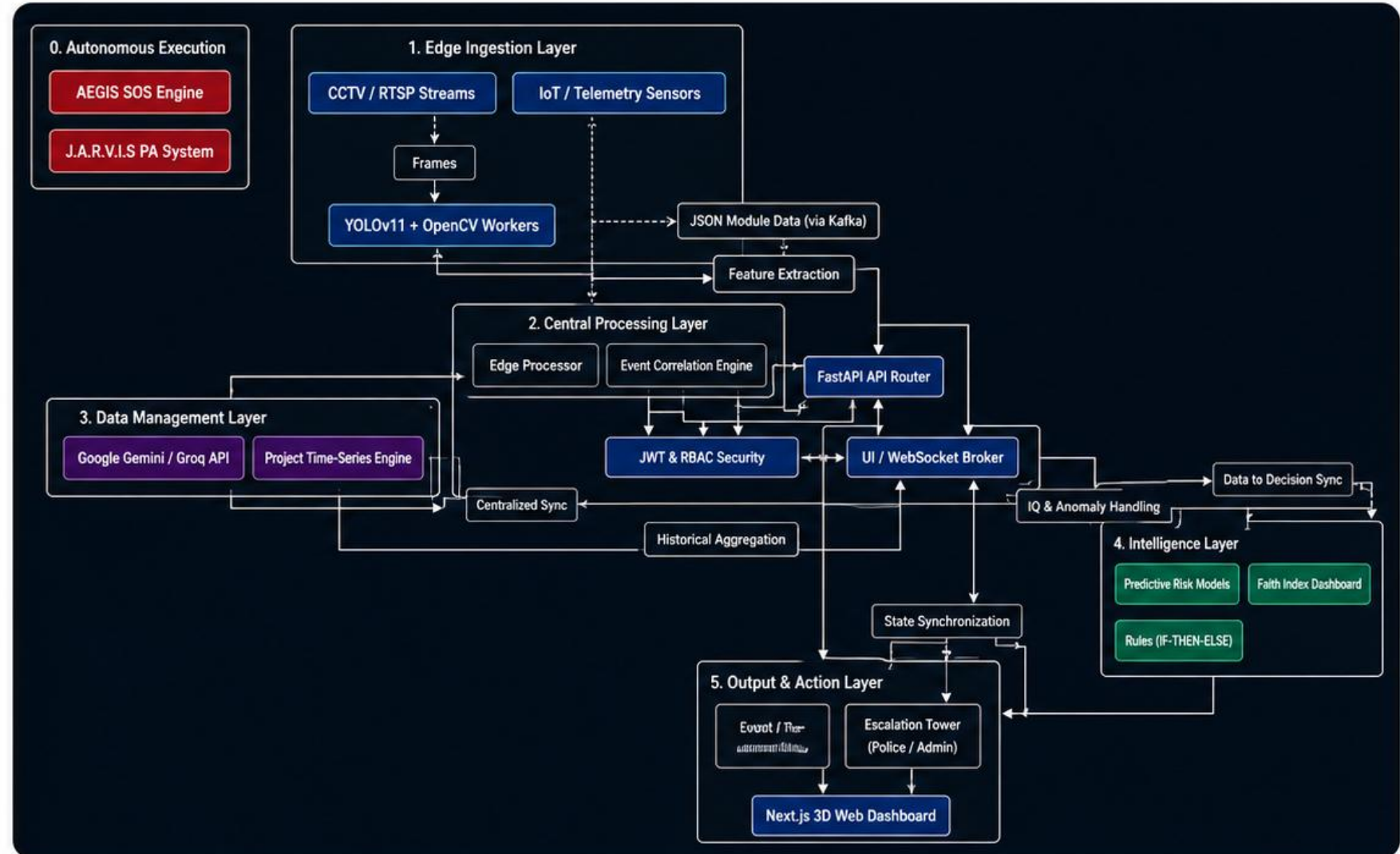


Real-Time Event Streaming

Instant alerts and dashboard synchronization



Processes **city-wide events in real time** while remaining modular, scalable, and deployment-ready.





TECHNOLOGY STACK

Programming Languages

Technology	Purpose
 Python 3.11+	AI Processing, Backend Services
 TypeScript	Frontend Logic & Type Safety
 SQL	Database Queries & Analytics

Frontend Technologies

Technology	Purpose
 Next.js 15 + React 19	Modern Dashboard Framework
 Tailwind CSS	Responsive UI Design
 Framer Motion	Interactive Animations
 React Three Fiber	3D Visualization Engine

Backend & APIs

Technology	Purpose
 FastAPI	High-Performance Async Backend
 JWT + RBAC	Authentication & Access Control
 WebSockets	Real-Time Communication
 SSE Streaming	Live Dashboard Updates
 SMTP Services	Emergency Email Alerts

AI & Machine Learning

Technology	Purpose
 YOLOv11	Real-time Object Detection
 OpenCV	Vision Processing & Analytics
 Google Gemini	Zero-Shot Intelligence & Reasoning
 Groq API	Randy AI & Fast Inference
 Prophet	Crowd & Traffic Forecasting

Data & Storage

Technology	Purpose
 PostgreSQL	Structured Data Storage
 Faiss Vector DB	Natural Language Video Search
 Event Logs	Historical Incident Records

Why Laminar is Feasible?

- ✓ Edge AI Processing
- ✓ Real-Time Event Streaming
- ✓ Low Bandwidth Architecture
- ✓ Modular & Scalable Design
- ✓ Production-Ready Technology Stack

FEATURES & IMPLEMENTATION



Public Safety & Emergency Intelligence

Feature	Implementation
 Crowd Intelligence	YOLOv11 Crowd Density Analysis
 Guardian Route	Safe Path Navigation + SOS Monitoring
 Sentinel Beacon	One-Touch Emergency Alert + GPS + Email
 AEGIS Protocol	Fall Detection & Medical Emergency Response
 Amber Protocol	Missing Person Detection & Tracking

- ✓ Unified AI Operating System
- ✓ Predictive + Reactive Intelligence
- ✓ Cross-System Decision Making
- ✓ Real-Time Emergency Coordination
- ✓ AI-Powered Incident Reconstruction


**11 Integrated
Modules • One
Unified Intelligence
Platform**



Smart City Operations

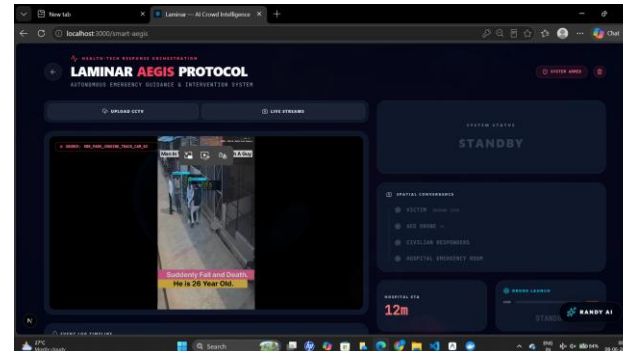
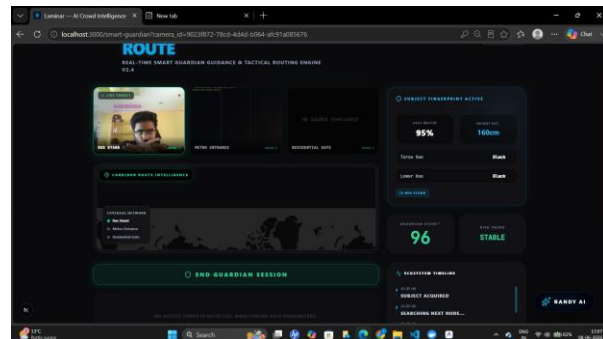
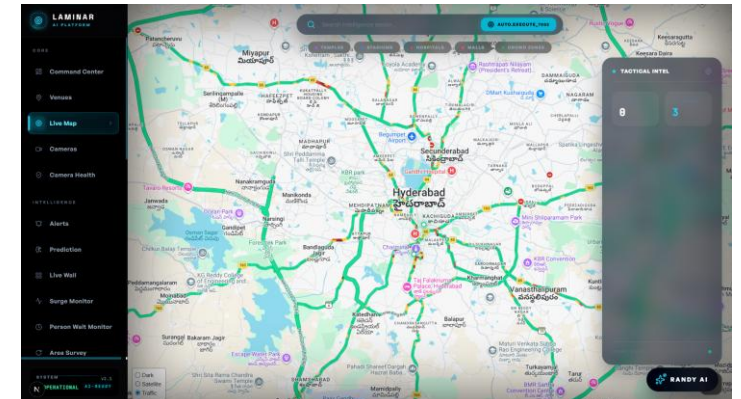
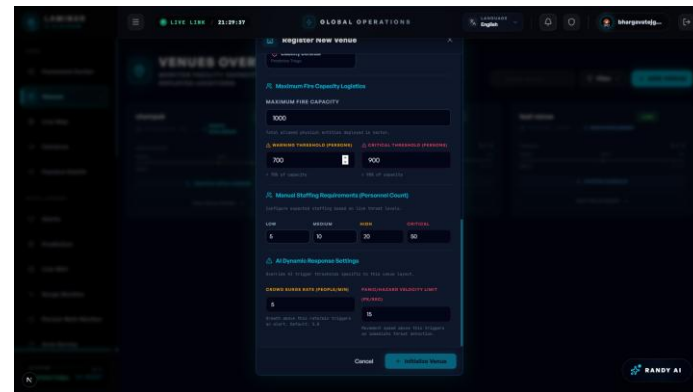
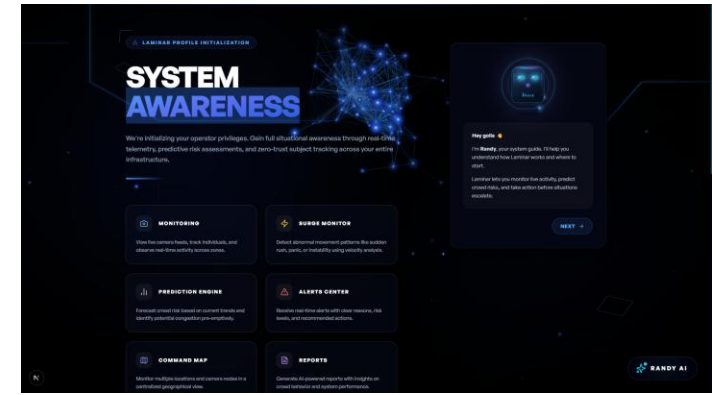
Feature	Implementation
 AI Green Wave	Emergency Vehicle Priority Routing
 Flood Intelligence	Waterlogging & Flood Risk Monitoring
 Resonance Bridge Intelligence	Structural Vibration Monitoring
 AI Video Search	Natural Language CCTV Search
 Randy AI	AI Investigation & Decision Support

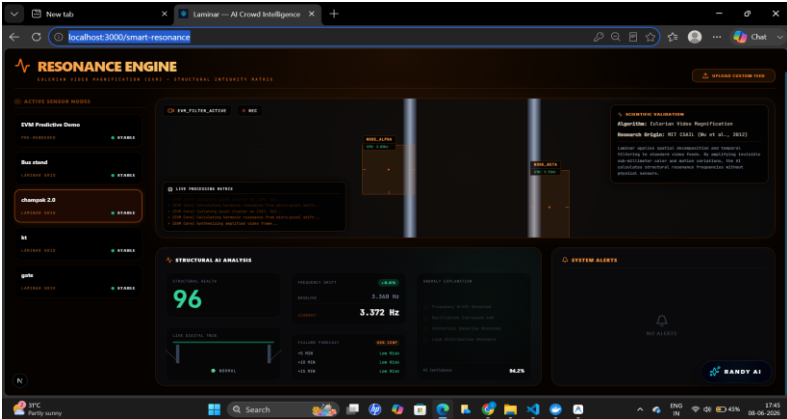
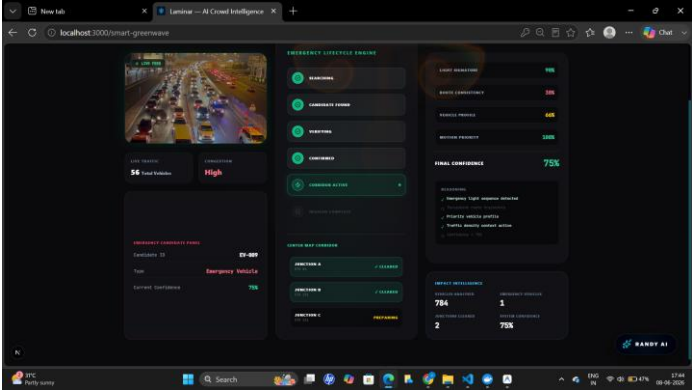
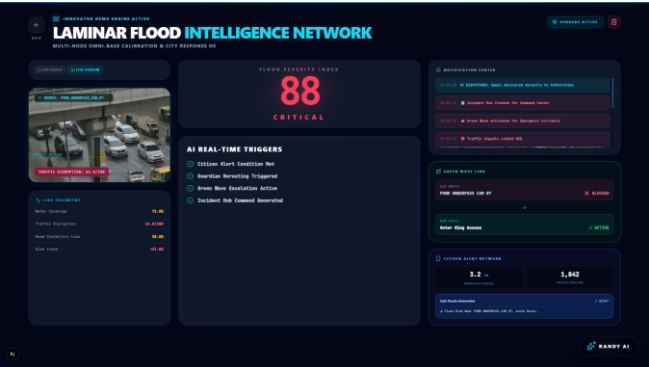
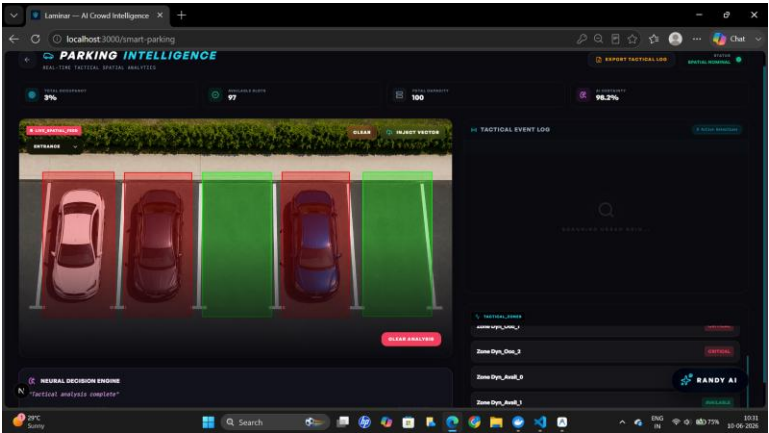
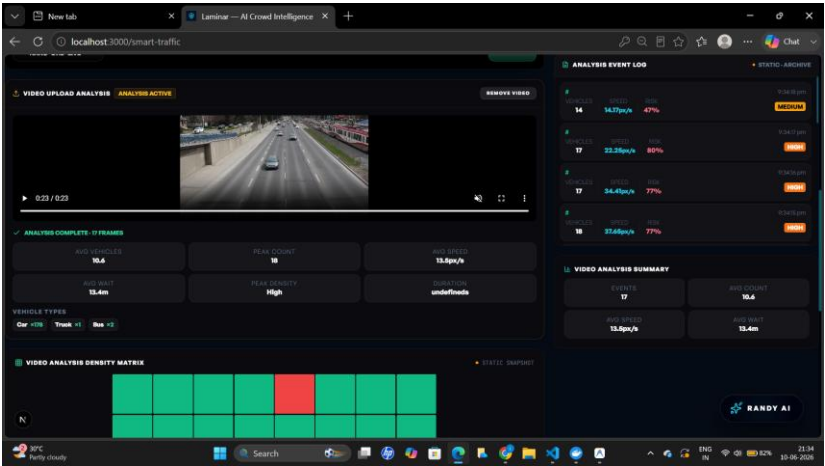
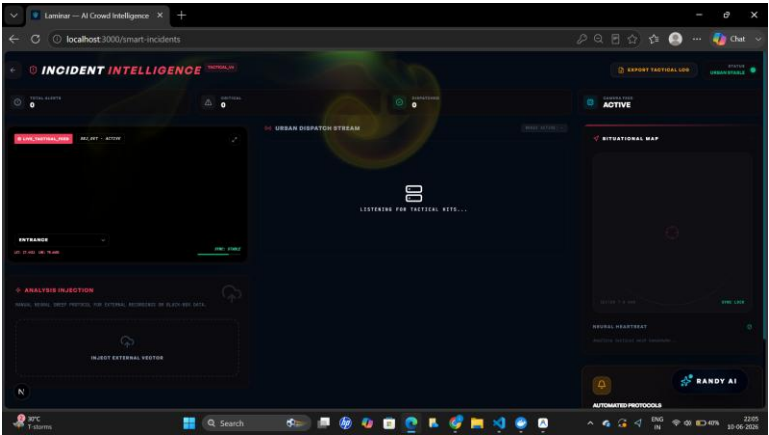
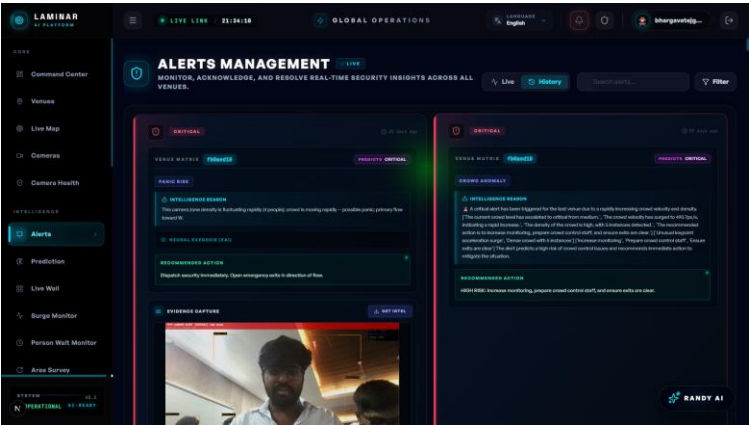


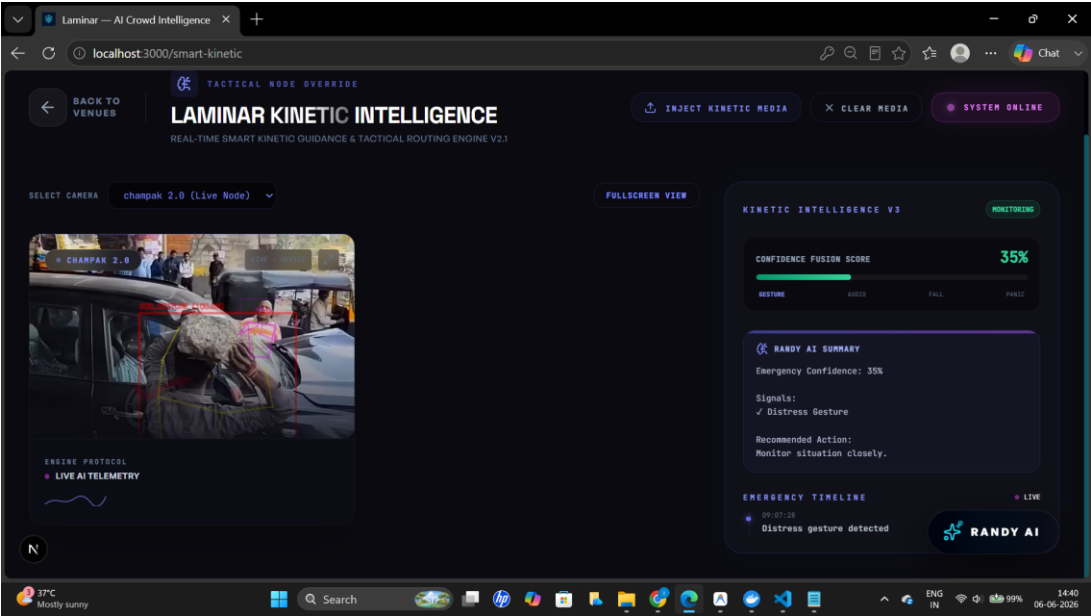
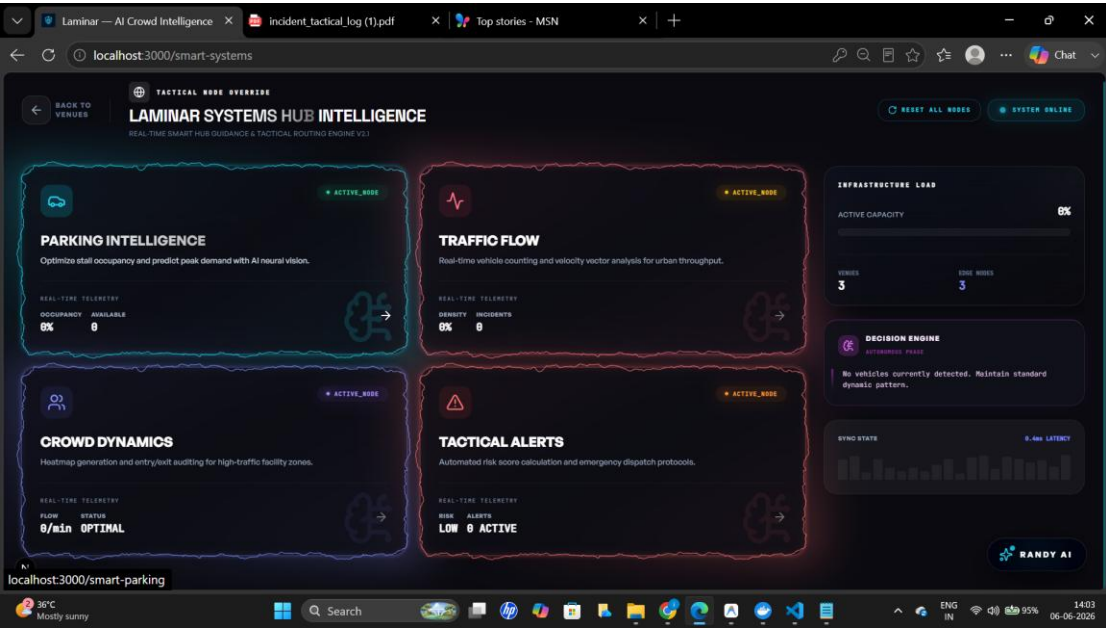
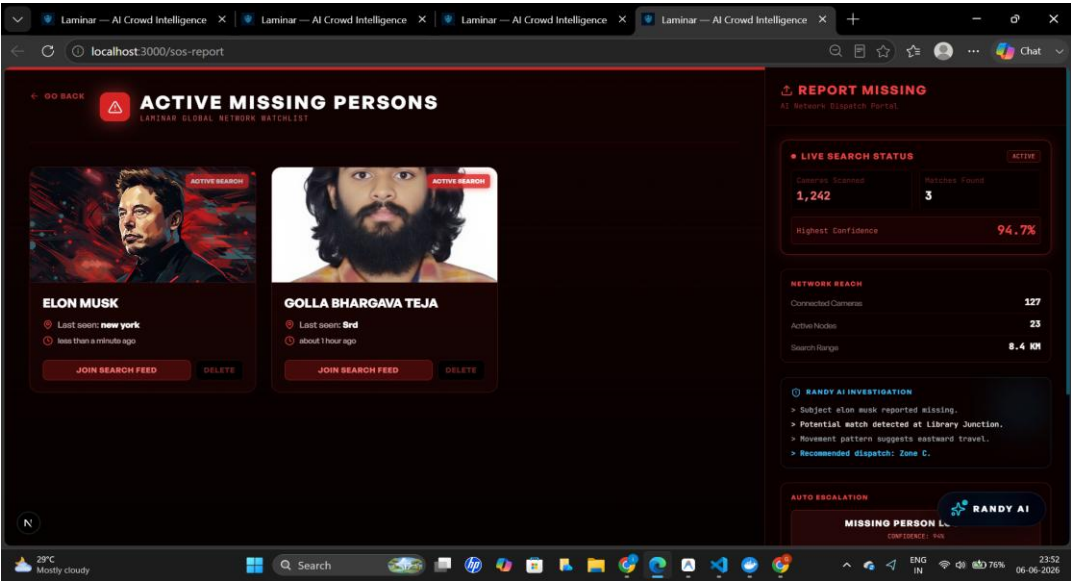
Incident Intelligence Layer

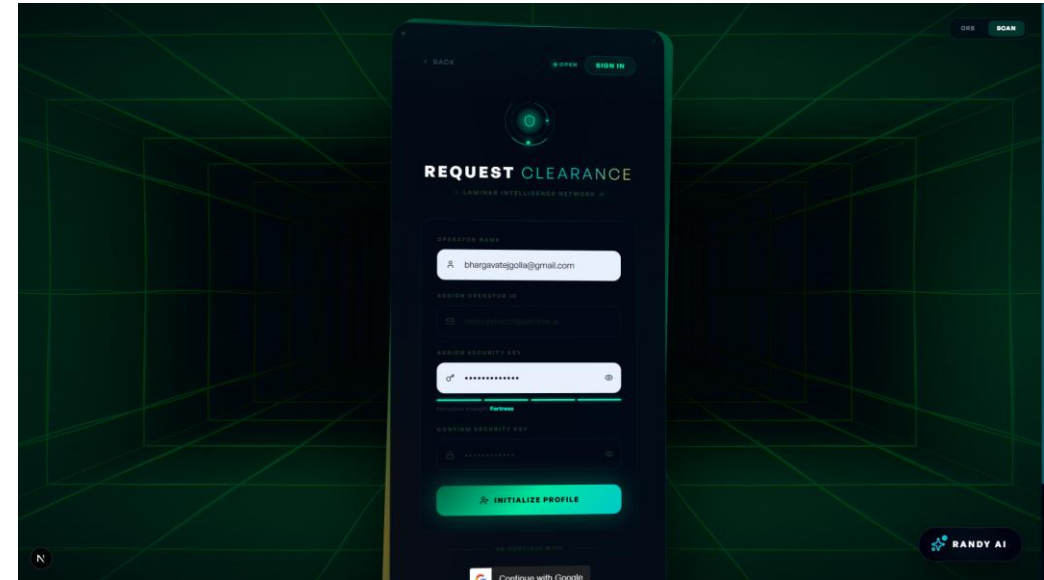
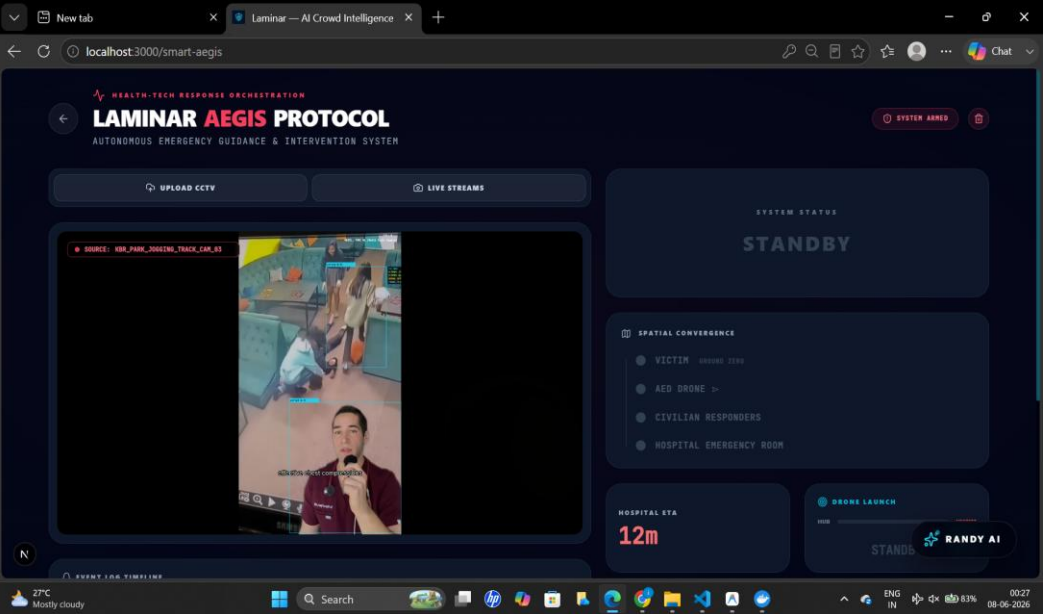
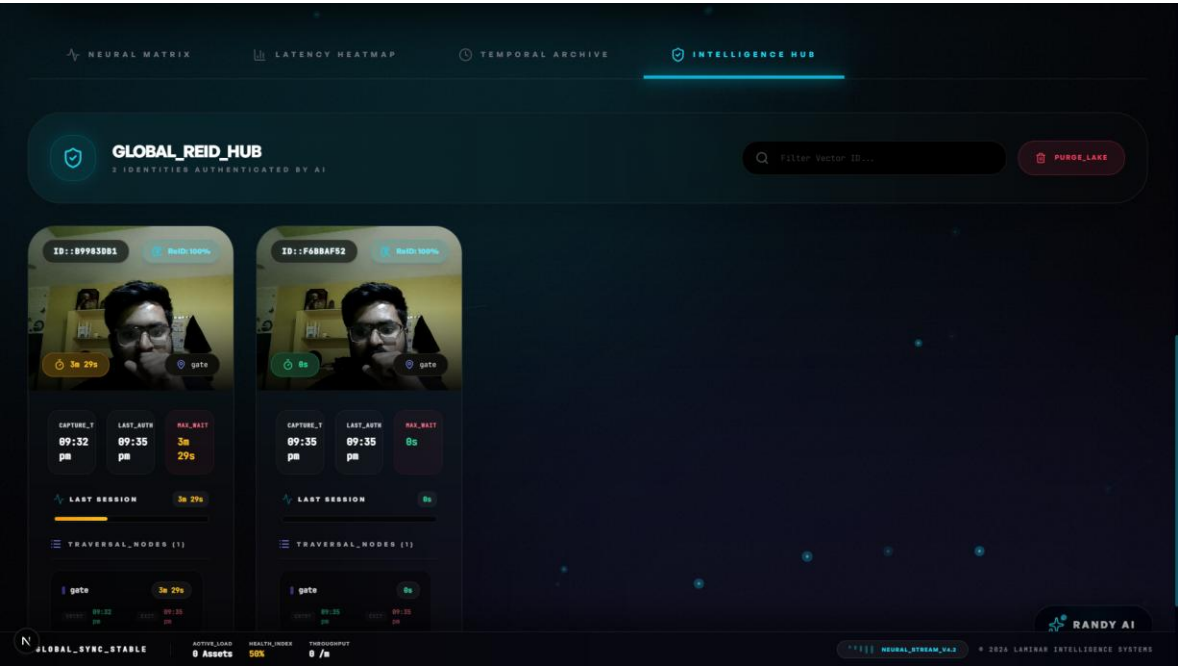
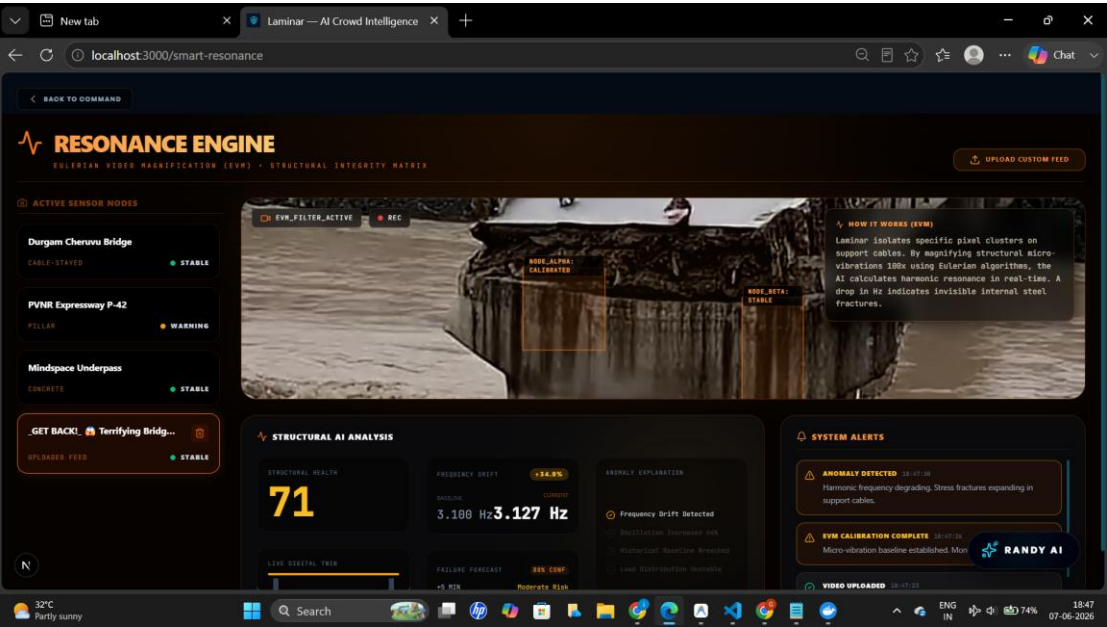
Feature	Implementation
 Liability Engine	AI Incident Reconstruction
 4D Playback Engine	Historical Event Replay
 Evidence Package Generator	Automated Case Dossier Creation
 Black Box Recorder	Forensic Audit Trail
 Cross-System Evidence Fusion	Multi-Source Intelligence Correlation

RESULTS /DEMO









FUTURE SCOPE & SCALABILITY



Smart City Expansion

Current: Campus / Public Venue Intelligence



Future Enhancements



Multi-City Command Centers



Integrated Traffic Networks



Public Safety Infrastructure



Smart Governance Systems



National Emergency Operations Systems



Why Laminar Scales

- ✓ Modular Architecture
- ✓ Edge AI Processing
- ✓ Cloud-Native Deployment
- ✓ API Driven Integrations
- ✓ Supports Thousands of Cameras & Sensors



Advanced AI Evolution



Future Enhancements:

- ✓ Digital Twin Cities
- ✓ Autonomous Incident Prediction
- ✓ Multi-Agent AI Coordination
- ✓ City-Wide Risk Forecasting
- ✓ Conversational Command Centers



Market & Application Scope

Sector	Application
Smart Cities	Public Safety & Operations
Universities	Campus Security
Transportation	Crowd & Traffic Management
Stadiums	Event Intelligence
Healthcare	Emergency Response
Industrial Zones	Infrastructure Monitoring



Laminar aims to become the **AI Operating System** that powers people, infrastructure, mobility, safety, and emergency services into a single **intelligent urban ecosystem**.



2026

Campus & Venue Deployment



2027

Smart Traffic Integration



2028

Multi City Operations



2030+

AI-Powered Autonomous City Network



CONCLUSION

What We Built:

- ✓ Unified AI-Powered Smart City Platform
- ✓ Real-Time Monitoring & Decision Support
- ✓ Predictive Risk Detection & Emergency Response
- ✓ AI Incident Reconstruction & Investigation
- ✓ Multi-Module Urban Intelligence Ecosystem

Key Achievements:

Achievement	Outcome
 AI Modules Integrated	11+
 Functional Dashboards	15+
 Emergency Intelligence	Implemented
 Incident Reconstruction	Implemented
 AI Video Search	Implemented
 Women Safety Features	Implemented

“Laminar transforms disconnected city systems into a **unified intelligence network** capable of observing, understanding, predicting, coordinating, and responding in real time.”



Smart City Expansion

Current: Campus / Public Venue Intelligence



Future Enhancements



Multi-City Command Centers



Integrated Traffic Networks



Public Safety Infrastructure



Smart Governance Systems



National Emergency Operations Systems



Advanced AI Evolution



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Digital Twin Cities



Autonomous Incident Prediction



Multi-Agent AI Coordination



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Conversational Command Centers



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Modular Architecture



Edge AI Processing



Cloud-Native Deployment



API-Driven Integrations



Supports Thousands of Cameras & Sensors

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Innovate for **India**. Build for **Tomorrow**.



THANK YOU!



FOR BEING A PART OF
PROJECT VIKSIT BHARAT 2026



INNOVATE.
SOLVE REAL WORLD
CHALLENGES.



COLLABORATE.
LEARN. GROW.
INSPIRE.



IMPACT.
BUILD SOLUTIONS
FOR A BETTER INDIA.



TRANSFORM.
IDEAS TODAY.
CHANGE TOMORROW.

Together, let's build a *Viksit Bharat*. Together, let's build the *Future*.

